

Measurement of Moisture and Dry Content in Paints with Mettler Toledo Halogen Moisture Analyzers

Ensuring Consistent Quality



Dry or moisture content play a critical role in both ensuring consistent quality and achieving the ideal viscosity of paints. In solvent-based paints, it is the volatile organic compounds (VOCs) that are of interest, and these are typically referred to as moisture. Whereas, in water-based paints and emulsions, the critical parameter is the dry content.

There are many types of paints and coatings. They are used in multiple industries, from home décor through to automobile and industrial applications. Measuring moisture and dry content in paints is governed by regulatory bodies around the world, based on industry, application, and type of paint formulation.

This application note describes how METTLER TOLEDO halogen moisture analyzers provide a fast, accurate and easy moisture measurement technique for multiple types of paints and coatings in line with industry regulatory standards.

Keywords or phrases:

Halogen moisture analysis, paints, coatings, quality control, volatile organic compounds (VOC), ASTM-compliant methods, in-process quality monitoring, Mettler Toledo

INTRODUCTION

Solvent-based paints are commonly used for coating metal surfaces, such as in the automobile industry, while water-based emulsions are used in coating applications in domestic and commercial environments, such as walls etc. However, in all applications, viscosity and spread are the key factors that determine the quality of paints and coatings.

This application note presents the fast and accurate measurement of moisture and dry content in paints using a halogen moisture analyzer from METTLER TOLEDO. The technique is applicable to moisture and dry content determination in diverse paint industry applications.

Paints, coatings, and varnishes

In multiple industries, painting is a means of preserving and protecting surfaces as well as decorating. Paints may be in a liquid, paste or solid powder form and are applied to the material or surface to a specific thickness and/or number of layers.

The term 'coating' is widely used to indicate the application of a material to a surface. 'Paint' generally indicates that a pigment has been used and this is often the distinguishing factor of this type of coating. There are many different types of paint, from water-based emulsions to solvent-based enamels and more. 'Varnishes' are defined as a lacquer, which is a type of coating that dries only by evaporation.

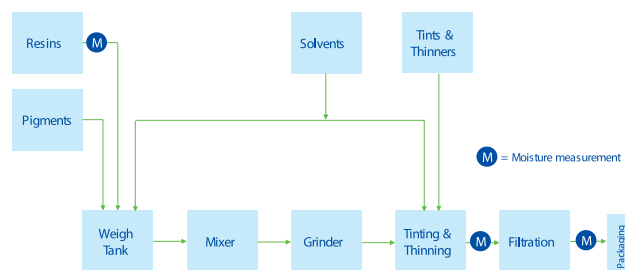


Figure 1: Flow diagram for manufacture of solvent-based paints

Moisture is the critical component

Irrespective of the type, the key ingredients, and the application, the most critical component in paints is 'moisture'. The moisture content typically comes from a carrier, which is normally a combination of different organic solvents, or water, in the case of emulsions used for domestic decorative and protective applications.

For solvent-based paints, moisture refers to the total volatile organic compounds (VOCs). In water-based paints, determining the solid content (dry content) is important. With both types, moisture measurement is critical to maintain quality and achieve the viscosity (thickness) that strikes the right balance between the ability to spread, percolate, and coat the surface with a specific thickness.

The overall consumption of paint for it to perform its function determines its commercial viability. This is related

to the cost of solvents, pigments, and resins, together with the number of coats required to paint a surface. Testing moisture content at key stages in the manufacturing process is therefore critical, including testing raw materials such as resins and binders, verifying viscosity during manufacture, and quality control and stability testing of the finished product (Figure 1).

Moisture testing regulations

The American Society for Testing and Materials (ASTM) is one of the global regulatory bodies that defines the quality testing for paint and coatings. ASTM: D2832–9211 and ASTM: D2369–1022 are related regulations for paint and coating-based applications and specify the drying oven method for moisture testing.

A halogen moisture analyzer is an alternative method, which is also based on thermogravimetric principles.

Moisture analyzers: fast, accurate and safe

One of the main advantages of using a moisture analyzer is that results are delivered in minutes instead of hours, enabling faster batch processing. A moisture analyzer can also be placed alongside production lines; the fast heat dissipation and the ability to withstand challenging operational environments make it a safe and practical solution, especially for solvent-based paint production.

MATERIAL AND METHODS: MEASURING MOISTURE CONTENT IN PAINT WITH THE HX204 MOISTURE ANALYZER

- METTLER TOLEDO HX204 halogen moisture analyzer.
- Sample: enamel paint.

1. Choose the 'Enamel' method Drying Program: Standard

- Drying Temperature: 140 °C
- Switch-off criterion: 3 (1 mg/50 s)
- Display unit: %MC



2. Weigh out around 2 g of the enamel paint onto a tared sample pan containing a glass fibre filter.
3. Press "Start Drying" to begin the measurement.
4. The result and the real-time drying curve are displayed.

RESULTS

Enamel is a solvent-based paint. The moisture content, predominantly VOCs, was measured using the HX204. The

average of three measurements was found to be 35.65 %MC. The result fulfilled the recommended maximum limit as per an internal company standard for the sample under study (max. 40 %MC).

The moisture content result guaranteed the quality in terms of flowability/thickness of the paint. With METTLER TOLEDO's recommended drying method, the results were obtained in around 9 minutes compared to 120 minutes using the drying oven method according to the same company's internal method of analysis.

Correlation HX204 versus Drying Oven

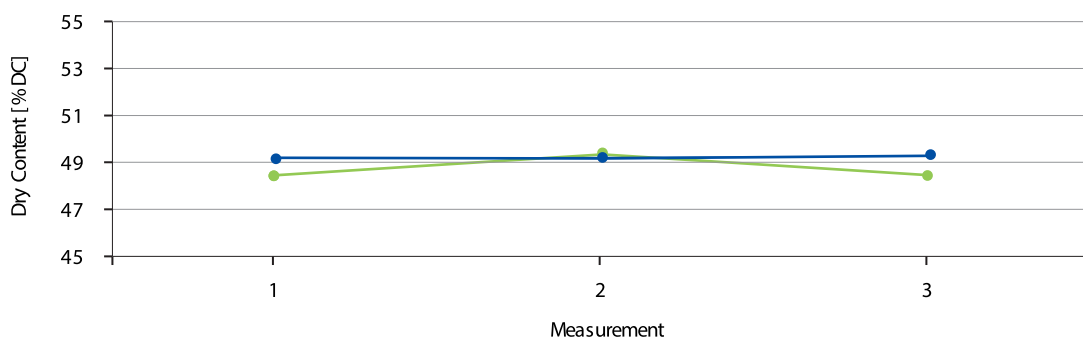
To prove the precision of the results from the HX204, a comparison study was performed with three different samples used in the paint industry. Results from the HX204 were compared to the drying oven method.

The study showed that the HX204 achieved highly repeatable results for all three samples, corresponding fully to those from the drying oven method, with results in minutes instead of hours.

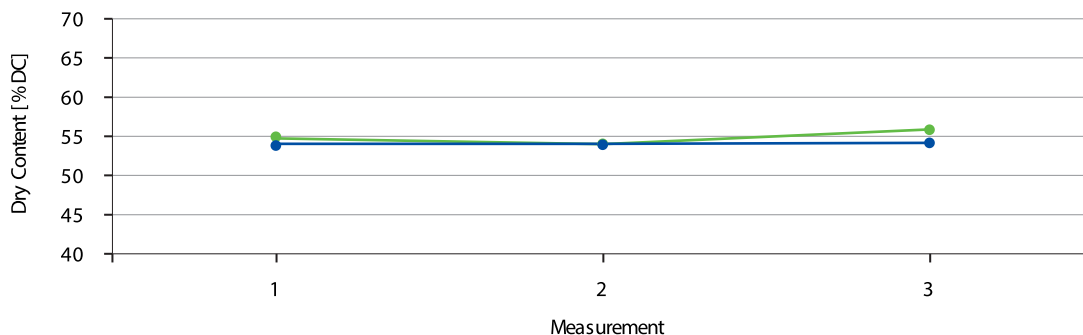
Paint Industry Samples	HX204			Drying Oven		
	Mean (%DC)	SD	Time (min)	Mean (%DC)	SD	Time (min)
Thinner (Water based)	49.18	0.24	6	48.75	0.42	120
Emulsion (Water based)	54.08	0.06	10	54.86	0.77	120
Enamel (Solvent based)	64.42	0.17	9	64.56	0.14	120

Table 1: Comparison of results of dry content determination using the HX204 (average of 3 measurements) versus the drying oven (average of 3 measurements). *For comparison purposes, dry content results are provided DC = dry content, SD = standard deviation.

Thinner (Water based)



Emulsion (Water based)



Enamel (Solvent based)*

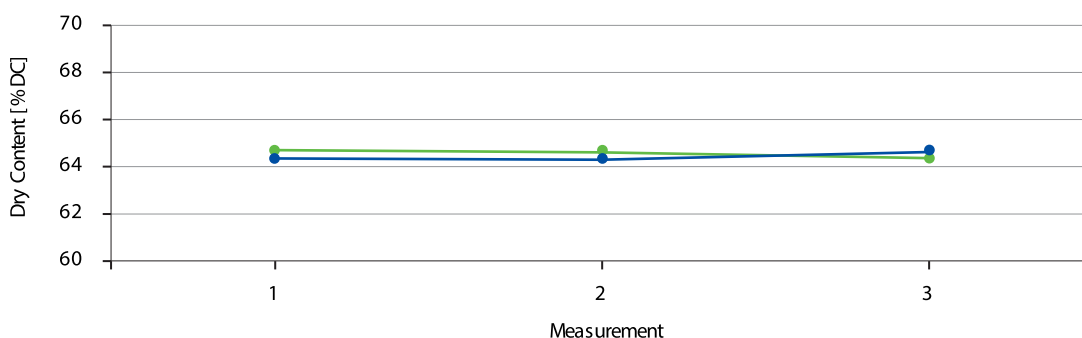


Figure 2: Comparison of results for thinner, emulsion, and enamel obtained with the HX204, halogen moisture analyzer (blue) and the drying oven (green).

CONCLUSION

The moisture and dry content in paints and coatings can be determined quickly and accurately using METTLER TOLEDO halogen moisture analyzers, with results corresponding fully to the drying oven reference method.

Thanks to appropriate drying method development, METTLER TOLEDO halogen moisture analyzers not only provide equivalent results to the drying oven reference method but offer a significantly shortened analysis time of just minutes rather than hours.

Moisture analyzers are therefore a valuable tool for providing fast and precise information on moisture and dry content. They can be used in parallel to the drying oven and can contribute significantly to ensuring quality in paint and related industries.

REFERENCES

- 1 ASTM D2832-92: Standard Guide for Determining Volatile and Nonvolatile Content of Paint and Related Coatings
- 2 ASTM D2369-10: Standard Test Method for Volatile Content of Coatings



HX204 Halogen Moisture Analyzer.